

SECMS

Student Enrollment & Course Management System
Built on the Salesforce Platform

Phase 7: Final Project Documentation

Project Name	SECMS
Platform	Salesforce Developer Edition (Lightning Experience)
Phase	7 of 8 - Project Documentation
Team Name / Members	Member 1: Vasanthakumar R Team Lead Member 2: Saravanan V Member 3: Syed Kutbudeen S Member 4: Raji P Member 5: Sriram S

Contents

- 1. Project Overview
- 2. Problem Statement & Objectives
- 3. Requirements
- 4. System Design
- 5. Implementation
- 6. Testing Summary
- 7. Results & Benefits
- 8. Future Scope
- 9. Conclusion
- 10. Project Repository Structure

1. Project Overview

SECMS (Student Enrollment & Course Management System) is a Salesforce Lightning application that manages students, courses, instructors, enrollments and fee payments in one place. It replaces scattered spreadsheets and manual follow-ups with connected records, automation, approvals, reports and a dashboard.

2. Problem Statement & Objectives

Problem: Student, course and fee information is often stored in separate files, so enrollments are slow, error-prone and hard to track.

Objectives:

- Centralize student, course, instructor, enrollment and fee data on Salesforce.
- Automate enrollment tasks using Flow and route special cases through an approval process.
- Protect data quality with validation rules.
- Give administrators real-time reports and a dashboard.

3. Requirements

3.1 Functional Requirements

ID	Requirement
FR1	Create and manage Student, Course and Instructor records
FR2	Enroll a student in a course and track enrollment status
FR3	Validate student email as an institutional email
FR4	Set enrollment date, assign instructor, send student email and create follow-up task automatically
FR5	Route enrollment requests to the Training Manager for approval

FR6	Track fee payment status
FR7	Provide reports and a dashboard for students, enrollments and fees

3.2 Non-Functional Requirements

- Usability: simple Lightning interface with tabs for each object.
- Security: role-based access through Salesforce profiles and permission sets.
- Reliability: data hosted on the Salesforce cloud platform.
- Scalability: new objects, fields and automations can be added without code.

4. System Design

4.1 Data Model

Object	Purpose	Relationships
Student	Student personal and contact details	Linked to Enrollment
Course	Course details, fee and duration	Linked to Enrollment and Instructor
Instructor	Instructor profile	Assigned to courses / enrollments
Enrollment	Connects a Student to a Course with status and date	Student + Course
Payment / Fees	Fee payment status for an enrollment	Linked to Enrollment

4.2 Automation Design

Component	Type	Purpose
Student_Email_Domain_Validation	Validation Rule	Blocks emails that are not institutional emails
Set Enrollment Date	Record-Triggered Flow	Sets the enrollment date on save
AutoAssign_Instructor	Record-Triggered Flow	Assigns an instructor automatically
Send Student Email	Record-Triggered Flow	Emails the student after enrollment
Create_Followup_Task	Record-Triggered Flow	Creates a follow-up task
Enrollment Approved Action	Record-Triggered Flow	Runs actions after approval
Approval Enrollment Request	Approval Process	Sends enrollment requests to the Training Manager

5. Implementation

The application was built on a Salesforce Developer Edition org using point-and-click tools: Object Manager, App Manager, Flow Builder, Approval Processes, Reports and Dashboards.

5.1 Objects

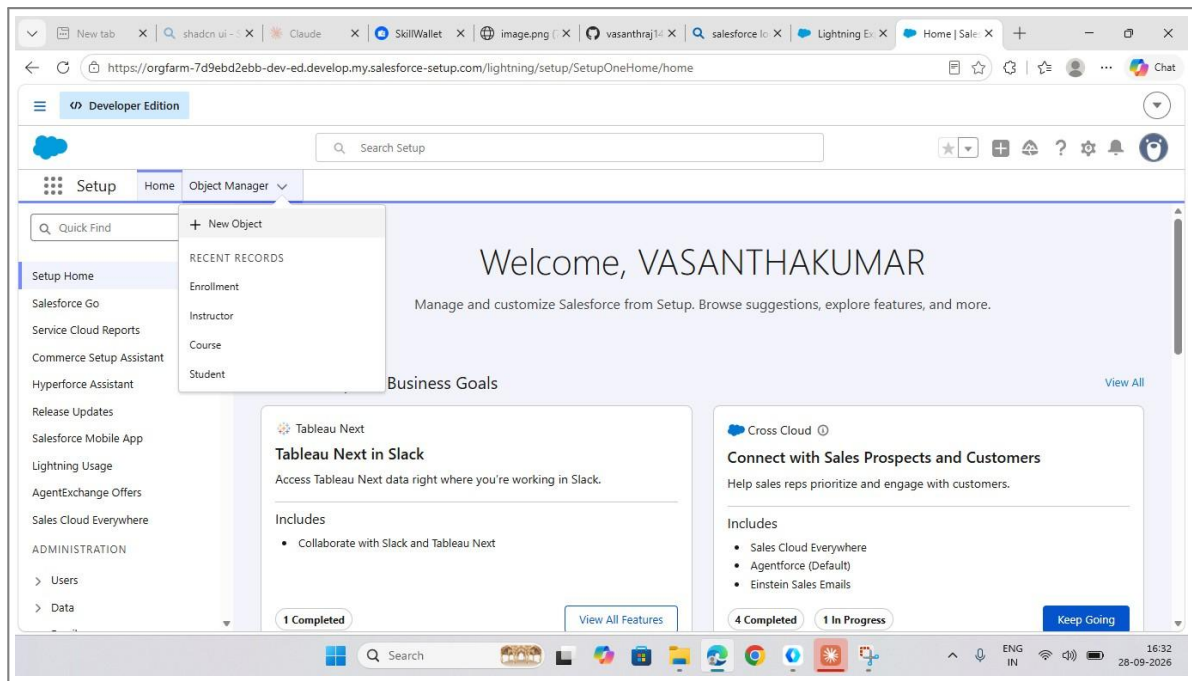


Figure 1: Custom objects Enrollment, Instructor, Course and Student in Object Manager

5.2 Lightning App

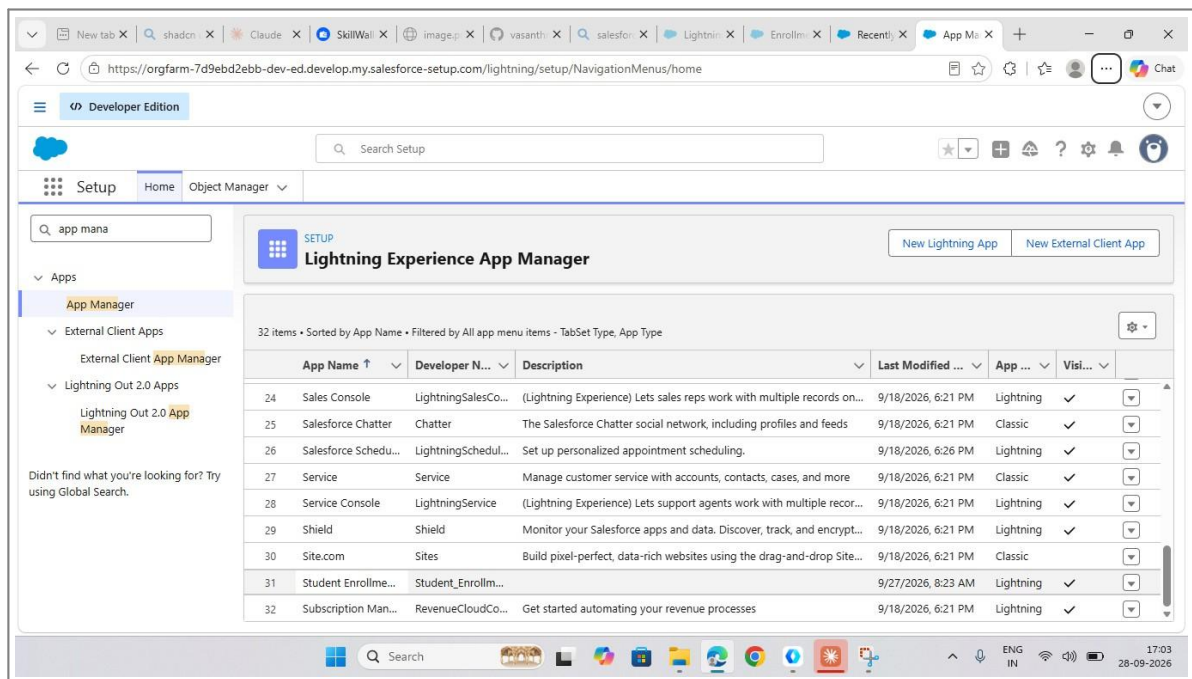


Figure 2: Student Enrollment Lightning app in App Manager

5.3 Validation Rule

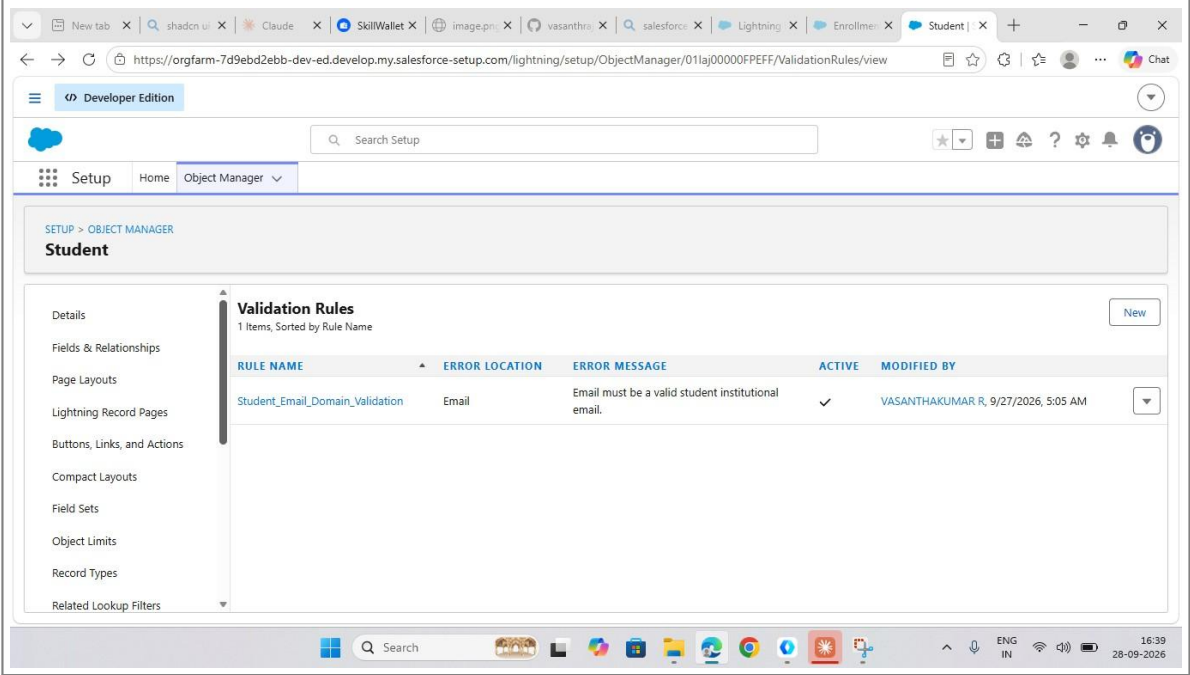


Figure 3: Student_Email_Domain_Validation rule (active)

5.4 Flow Automation

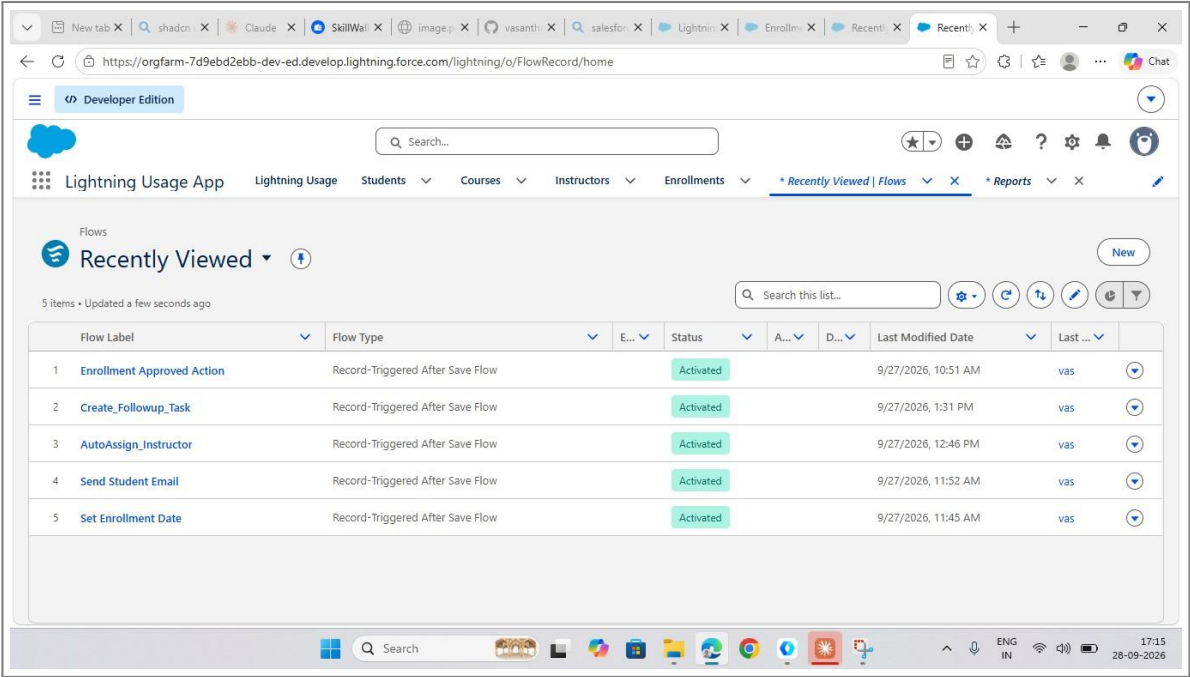
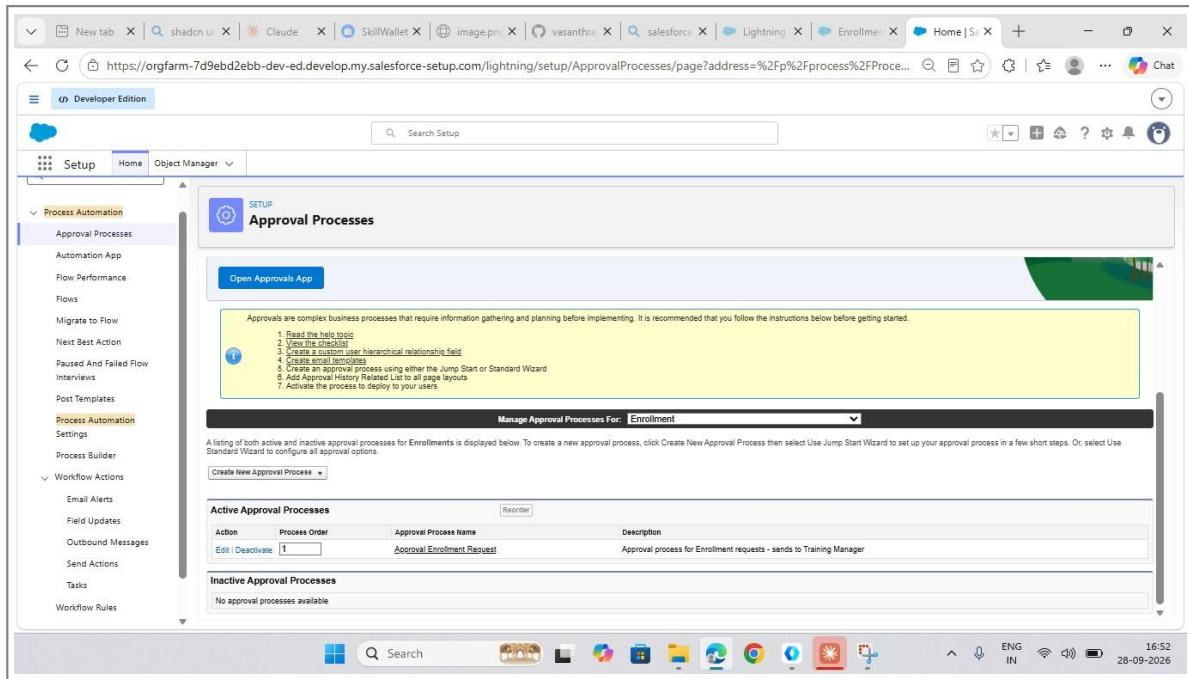


Figure 4: Five activated record-triggered flows



5.5 Approval Process

Figure 5: Approval Enrollment Request process for the Enrollment object

5.6 Reports

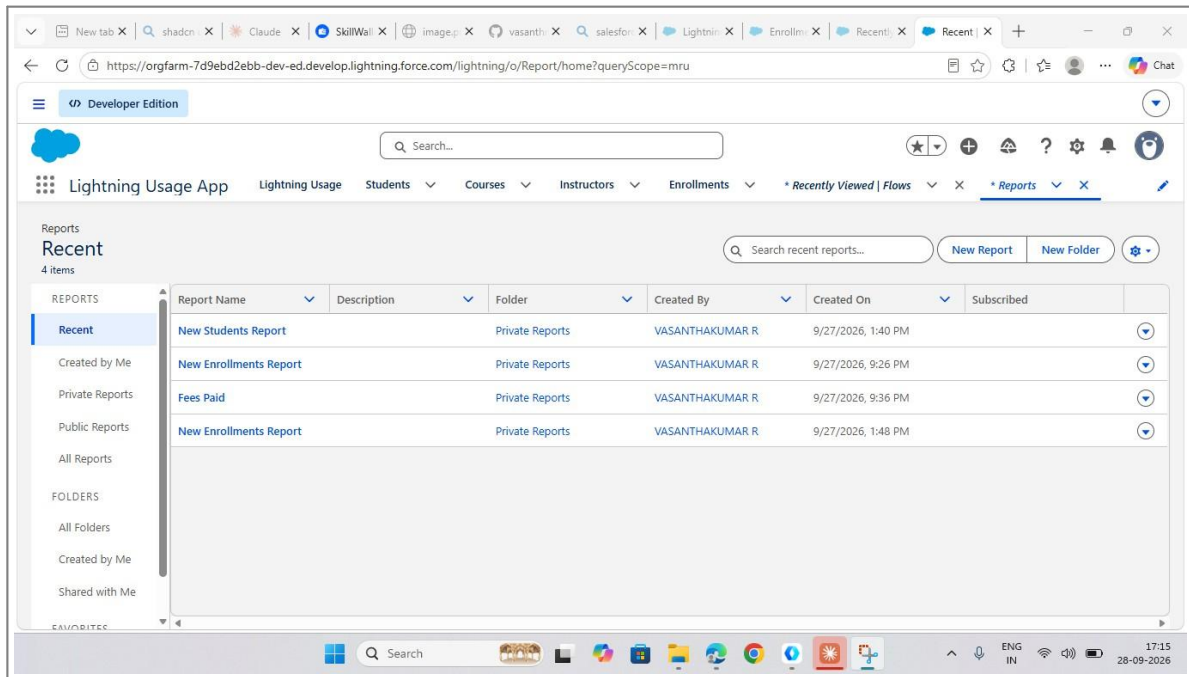


Figure 6: New Students, New Enrollments and Fees Paid reports

5.7 Dashboard

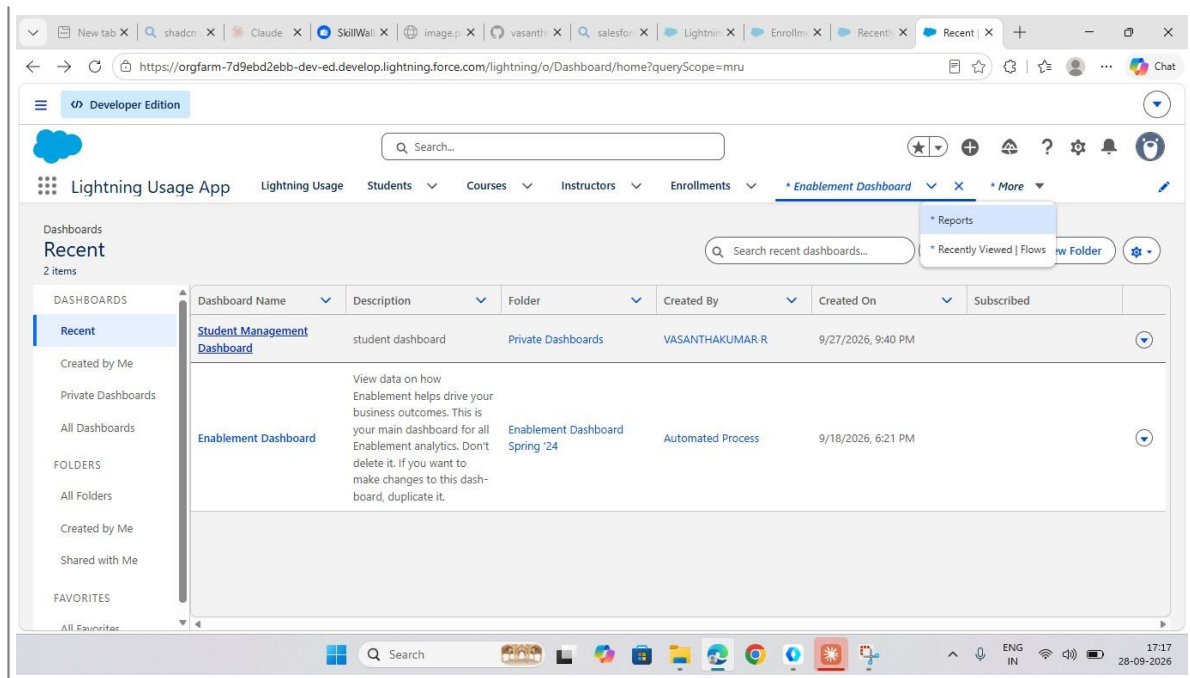


Figure 7: Student Management Dashboard

6. Testing Summary

Functional testing covered records, validation, flows, approval process, reports and dashboard. Details are in the Phase 6 Testing Report.

Total Test Cases	Passed	Failed	Pass Rate
16	16	0	100%

7. Results & Benefits

- One connected system for students, courses, instructors, enrollments and fees.
- Less manual work because dates, instructor assignment, emails and tasks are automated.
- Better data quality through validation rules.
- Controlled enrollment through an approval process.
- Quick insight through reports and a dashboard.

8. Future Scope

- Experience Cloud student portal for self-service enrollment.
- Online payment gateway integration.
- Attendance and grade tracking objects.
- Mobile access through the Salesforce mobile app.
- Einstein / AI-based course recommendations.

9. Conclusion

SECMS shows how Salesforce can be used to build a complete student enrollment and course management solution without custom code. The project delivered structured data, automation, approvals and analytics, and it can be extended easily as requirements grow.

10. Project Repository Structure

Folder	Contents
01-Brainstorming-and-Ideation	Ideation PDF and diagram
02-Requirement-Analysis	Requirement analysis PDF
03-Project-Design	Design PDF, architecture and ER diagrams
04-Project-Planning	Project plan PDF and timeline
05-Project-Development	Salesforce screenshots
06-Project-Testing	Testing report and test screenshots
07-Project-Documentation	This document
08-Project-Demonstration	Demo script and video link